

SMOTE-Tomek

SMOTE-Tomek is a hybrid machine learning technique for handling imbalanced datasets, combining SMOTE (Synthetic Minority Over-sampling Technique) for oversampling the minority class and Tomek Links for undersampling by removing noisy and overlapping data points. First, SMOTE creates synthetic minority-class samples, potentially causing class overlap, then Tomek Links identify and remove pairs of nearest, but opposing-class, samples, enhancing class separation and improving model performance.

How SMOTE-Tomek Works

1. **SMOTE (Synthetic Minority Over-sampling Technique):**
 - This oversampling technique generates new, synthetic data points for the minority class.
 - It does this by interpolating between existing minority class samples, creating a more balanced class distribution.
2. **Tomek Links (Undersampling):**
 - After SMOTE, the synthetic samples might cause class clusters to "invade" each other's space, leading to noise or overlapping data.
 - A Tomek Link is a pair of instances from different classes that are the closest neighbors to each other.
 - The Tomek Links method then identifies and removes these "linked" samples, especially those near the decision boundaries, thereby increasing the separation and clarity between the classes.

Purpose of SMOTE-Tomek

- **Addresses Class Imbalance:**The primary goal is to balance datasets where one class has significantly fewer samples than others.
- **Removes Noise and Overlap:**It helps to clean the data by removing noisy or ambiguous samples that are difficult for a classifier to categorize accurately.
- **Improves Classifier Performance:**By creating a cleaner, better-defined decision boundary, SMOTE-Tomek can lead to improved accuracy and, crucially, better recall for the minority class.